CARBOplatin (AUC 5) and 5-Fluorouracil 1000mg/m^2/day Therapy – 28 day cycle

INDICATIONS FOR USE:

INDICATION	ICD10	Regimen Code	HSE approved reimbursement status*
Adjuvant chemotherapy following chemoradiation treatment in patients with locally advanced stage III or IV nasopharyngeal carcinoma where CISplatin is contraindicated or not tolerated	C11	00552a	N/A

*This applies to post 2012 indications

TREATMENT:

The starting dose of the drugs detailed below may be adjusted downward by the prescribing clinician, using their independent medical judgement, to consider each patients individual clinical circumstances.

CARBOplatin[i] is administered on Day 1 and 5-Fluorouracil is administered on days 1-4 of a 28 day cycle for up to 3 cycles until disease progression or unacceptable toxicity develops.

Adjuvant treatment should begin 4 weeks after completion of **NCCP Regimen 00419: CARBOplatin (AUC 2) Weekly with Radiotherapy (RT)**

Facilities to treat anaphylaxis MUST be present when systemic anti-cancer therapy (SACT) is administered.

Admin. Order	Day	Drug	Dose	Route	Diluent & Rate	Cycle
1	1	CARBOplatin	AUC 5	IV infusion	500mL glucose 5% over 30 minutes	1-3
2	1-4	[a,b]5-Fluorouracil	1000mg/m^2/day	Continuous IV infusion	Over 96 hours in 0.9% NaCl	1-3
[a]Alternatively can be administered at 1000mg/m^2 in 1000mL 0.9% NaCl as an intravenous infusion over 22 hours on days 1-4 for a total dose of 4000mg/m2 over 96 hours.						
[b]See dose modifications section for patients with identified partial dihydropyrimidine dehydrogenase (DPD) deficiency						

Note: Administration volumes and fluids have been standardised to facilitate electronic prescribing system builds.

CARBOplatin dose:

The dose in mg of CARBOplatin to be administered is calculated as follows:

(mg) = target AUC (mg/mL x minute) x (GFR mL/minute +25)

- **Measured GFR** (e.g. nuclear renogram) is preferred whenever feasible
- **Estimation of GFR (eGFR)** can be done by using the Wright formula or using the Cockcroft and Gault formula to measure creatinine clearance
- The GFR used to calculate the AUC dosing should not exceed 125mL/minute
- For obese patients and those with a low serum creatinine, for example, due to low body weight or post-operative asthenia, estimation using formulae may not give accurate results; measured GFR is recommended

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- Where obesity (body mass index [BMI] ≥ 30 kg/m2) or overweight (BMI 25-29.9) is likely to lead to an overestimate of GFR and isotope GFR is not available the use of the adjusted ideal body weight in the Cockcroft and Gault formula may be considered
- Where serum creatinine is less than 63 micromol/L, the use of a creatinine value of 62micromol/L or a steady pre-operative creatinine value may be considered
- These comments do not substitute for the clinical judgement of a physician experienced in prescription of CARBOplatin

WRIGHT FORMULA

There are two versions of the formula depending on how serum creatinine values are obtained, by the kinetic Jaffe method or the enzymatic method. The formula can be further adapted if covariant creatine kinase (CK) values are available (not shown).

1. *SCr measured using enzymatic assay.*

$$\text{GFR (mL/minute)} = \frac{(6230 - 32.8 \times \text{Age}) \times \text{BSA} \times (1 - 0.23 \times \text{Sex})}{\text{SCr (µmol/min)}}$$

2. *SCr measured using Jaffe assay*

$$\text{GFR (mL/minute)} = \frac{(6580 - 38.8 \times \text{Age}) \times \text{BSA} \times (1 - 0.168 \times \textit{Sex})}{\text{SCr (µmol/min)}}$$

Key: Sex = 1 if female, 0 if male; Age in years; BSA= DuBois BSA

COCKCROFT-GAULT FORMULA

$$\text{GFR (mL/minute)} = \frac{S \times (140 - \text{age in years}) \times \text{wt (kg)}}{\text{serum creatinine (micromol/L)}}$$

S= 1.04 for females and 1.23 for males

ELIGIBILITY:

- Indications as above following recovery from **NCCP Regimen 00417: CARBOplatin (AUC 2) Weekly with Radiotherapy (RT)**

EXCLUSIONS:

- Hypersensitivity to *CARBOplatin, 5-Flourouracil or any of the excipients
  *If it is felt that the patient may have a major clinical benefit from CARBOplatin, it may in exceptional circumstances be feasible to rechallenge a patient with a prior mild hypersensitivity reaction e.g. using a desensitisation protocol, but only with immunology advice, premedication as advised, and a desensitisation protocol under carefully controlled conditions with resuscitation facilities available and medical and/or ITU/ HDU supervision (6).
- Known complete dihydropyrimidine dehydrogenase (DPD) deficiency

PRESCRIPTIVE AUTHORITY:

The treatment plan must be initiated by a Consultant Medical Oncologist.

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TESTS:

Baseline tests:

- FBC, renal and liver profile
- Audiology referral as clinically indicated
- Isotope GFR measurement (preferred) or GFR / Cr Clearance estimation
- DPD testing prior to first treatment with 5-Fluorouracil using phenotype and/or genotype testing unless patient has been previously tested
    - In patients with moderate or severe renal impairment, blood uracil levels used for dihydropyrimidine dehydrogenase (DPD) phenotyping should be interpreted with caution, as impaired kidney function can lead to increased uracil blood levels. Consequently, there is an increased risk for incorrect diagnosis of DPD deficiency, which may result in under dosing of 5-Fluorouracil or other fluoropyrimidines, leading to reduced treatment efficacy. Genotype testing for DPD deficiency should be considered for patients with renal impairment.

Regular tests:

- FBC, renal and liver profile prior to each cycle
- Assessment of hands and feet for palmar-plantar erythrodysaesthesia (PPE) as clinically indicated

Disease monitoring:

Disease monitoring should be in line with the patient's treatment plan and any other test/s as directed by the supervising Consultant.

DOSE MODIFICATIONS:

- Any dose modification should be discussed with a Consultant
- Consider a reduced starting dose in patients with identified partial DPD deficiency
    - Initial dose reduction may impact the efficacy of treatment
    - In the absence of serious toxicity, subsequent doses may be increased with careful monitoring

Haematological:

Table 1: Dose modification for haematological toxicity

ANC ($x10^9$/L)		Platelets ($x10^9$/L)	Recommended Dose
≥1	and	≥100	100% dose
0.5-0.99	or	50-99	Delay treatment until recovery
<0.5	or	<50	Delay treatment until recovery and consider reducing CARBOplatin and 5-Fluorouracil by 25% for subsequent cycles
Febrile neutropenia			

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Renal and Hepatic Impairment:

Table 2: Recommended dose modification in renal and hepatic impairment

Drug	Renal Impairment	Hepatic Impairment	
CARBOplatin[a]	***See below**	No dose modification required	
5-Flourouracil[b]	No need for dose adjustment is expected	Mild and moderate (without renal impairment):	No need for dose adjustment is expected
	Haemodialysis: No need for dose adjustment is expected	Severe:	Not recommended

[a]Renal – Content agreed by clinicians for use across NCCP regimens, Hepatic – Giraud et al 2023
[b]Renal and hepatic dose modifications as per Giraud et al 2023

*Renal dysfunction and CARBOplatin:

- Patients with creatinine clearance values of < 60mL/minute are at greater risk to develop myelosuppression
- In case of GFR ≤ 20mL/minute CARBOplatin should not be administered at all
- If Cockcroft & Gault or Wright formula are used, the dose should be calculated as required per cycle based on a serum creatinine obtained within 48 hrs of drug administration
- If isotope GFR is used, the dose should remain the same provided the serum creatinine is ≤110% of its value at the time of the isotope measurement. If the serum creatinine is higher than this, consideration should be given to remeasuring the GFR or to recalculating using Cockcroft & Gault or Wright formulae

Management of adverse events:

Table 1: Dose Modification for Adverse Events

Adverse reactions	Recommended dose modification
Dysphagia or stomatitis:	
Grade 0-2	100%
Grade 3	Delay until improvement and proceed at 75-100%
Grade 4	Discontinue
Weight loss from baseline:	
≤10%	100%
>10%	75% if hyperalimentation instituted, otherwise discontinue
Palmar-plantar erythrodysaesthesia:	
Grade 0-2	100%
Grade 3	75%

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SUPPORTIVE CARE:

EMETOGENIC POTENTIAL:

- As outlined in NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting - Available on the NCCP website

CARBOplatin: High **(Refer to local policy)**
5-Fluorouracil: Low **(Refer to local policy)**

For information:
Within NCIS regimens, antiemetics have been standardised by Medical Oncologists and Haemato-oncologists and information is available in the following documents:

- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Medical Oncology) - Available on the NCCP website
- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Haemato-oncology) - Available on the NCCP website

PREMEDICATIONS: No specific recommendations

- **OTHER SUPPORTIVE CARE**:
- Anti-diarrhoeal treatment (Refer to local policy).
- Management of PPE (refer to local policy)

ADVERSE EFFECTS:

- Please refer to the relevant Summary of Product Characteristics for details.

REGIMEN SPECIFIC COMPLICATIONS:

CARBOplatin

- **Hypersensitivity:** Reactions to CARBOplatin may develop in patients who have been previously exposed to platinum therapy. However allergic reactions have been observed upon initial exposure to CARBOplatin.
- **Neurotoxicity and ototoxicity**: Neurological evaluation and an assessment of hearing should be performed on a regular basis, especially in patients receiving high dose CARBOplatin. Neurotoxicity, such as parasthesia, decreased deep tendon reflexes, and ototoxicity are more likely seen in patients previously treated with cisplatin, other platinum treatments and other ototoxic agents. Frequency of neurologic toxicity is also increased in patients older than 65 years.

5-Fluorouracil

- **Myocardial ischaemia and angina:** Cardiotoxicity is a serious complication during treatment with fluorouracil. Patients, especially those with a prior history of cardiac disease or other risk factors, treated with 5-Fluorouracil, should be carefully monitored during therapy.
- **DPD deficiency:** DPD is an enzyme encoded by the DPYD gene which is responsible for the breakdown of fluoropyrimidines. Patients with DPD deficiency are therefore at increased risk of fluoropyrimidine-related toxicity, including for example stomatitis, diarrhoea, mucosal inflammation, neutropenia and neurotoxicity. Treatment with 5-Fluorouracil, capecitabine or tegafur-containing medicinal products is contraindicated in patients with known complete DPD deficiency. Consider a reduced starting dose in patients with identified

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partial DPD deficiency. Initial dose reduction may impact the efficacy of treatment. In the absence of serious toxicity, subsequent doses may be increased with careful monitoring. Therapeutic drug monitoring (TDM) of 5-Fluorouracil may improve clinical outcomes in patients receiving continuous 5-Fluorouracil infusions.

- **Hand-foot syndrome (HFS): HFS,** also known as palmar-plantar erythrodysaesthesia (PPE), has been reported as an unusual complication of high dose bolus or protracted continuous therapy for 5-Fluorouracil.

DRUG INTERACTIONS:

- Current SmPC and drug interaction databases should be consulted for information.

REFERENCES:

1. Chitapanarux, I., V. Lorvidhaya, P. Kamnerdsupaphon, et al. Chemoradiation comparing cisplatin versus CARBOplatin in locally advanced nasopharyngeal cancer: randomised, non-inferiority, open trial. Eur J Cancer 2007;43(9):1399-1406.#
2. Langendijk, J. A., C. R. Leemans, J. Buter, et al. 2004. "The additional value of chemotherapy to radiotherapy in locally advanced nasopharyngeal carcinoma: a meta-analysis of the published literature." J.Clin Oncol. 22(22):4604-461
3. Ouyang, P. Y., Xie, C. & Mao, Y. P. et al. 2013 "Significant efficacies of neoadjuvant and adjuvant chemotherapy for nasopharyngeal carcinoma by meta-analysis of published literature-based randomized, controlled trials." Ann Oncol 24(8):2136-46.
4. Appropriate chemotherapy dosing for obese adult patients with cancer: American Society of Clinical Oncology Clinical Practice Guideline. J Clin Oncol 2012; 30(13):1553-1561.
5. Ekhart C, Rodenhuis S et al. Carboplatin dosing in overweight and obese patients with normal renal function, does weight matter? Cancer Chemother Pharmacol 2009;64:115-122.
6. NCCN Guidelines Version 3.2017 Epithelial Ovarian Cancer/Fallopian Tube Cancer/Primary Peritoneal Cancer
7. Giraud E L, Lijster B D, et al. Dose recommendations for anticancer drugs in patients with renal or hepatic impairment: an update. Available at: https://pubmed.ncbi.nlm.nih.gov/37269847/
8. NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting. V6 2025. Available at: https://www.hse.ie/eng/services/list/5/cancer/profinfo/chemoprotocols/nccp-classification-document-for-systemic-anti-cancer-therapy-sact-induced-nausea-and-vomiting.pdf
9. CARBOplatin 10mg/mL Summary of Product Characteristics. Last updated 18/07/25. Accessed January 2026.Available at: https://assets.hpra.ie/products/Human/27514/Licence_PA2315-080-001_04092024114146.pdf
10. Fluorouracil 50mg/mL injection Summary of Product Characteristics. Last updated 29/10/2025. Accessed Jan 2026. Available at: https://assets.hpra.ie/products/Human/27540/Licence_PA2315-091-001_05032025160439.pdf

Version	Date	Amendment	Approved By
1	11/02/2019		Prof Maccon Keane
2	12/02/2020	Update of exclusions	Prof Maccon Keane
3	20/08/2020	Updated exclusion criteria, baseline testing, dose modifications and adverse events with respect to DPD deficiency as per DHPC from HPRA June 2020 Updated Adverse events regarding palmar-plantar erythrodysaesthesia	Prof Maccon Keane

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4	03/02/2021	Amended emetogenic potential	Prof Maccon Keane
5	10/10/2022	Updated CARBOplatin infusion time. Updated standard wording for CARBOplatin dosing, renal dysfunction and creatinine value. Updated baseline tests.	Prof Maccon Keane
5a	21/11/2023	Formatting changes and grammatical corrections.	NCCP
5b	13/03/2025	Additional wording added to the baseline testing section.	NCCP
6	12/03/2026	Regimen reviewed. Updated regular tests. Updated renal and hepatic dose modifications table to align with Giraud et al 2023. Addition of Table 3. Updated supportive care section. Updated regimen throughout in line with NCCP standardisation.	Dr Colm Mac Eochagáin

Comments and feedback welcome at oncologydrugs@cancercontrol.ie.

[i] This indication is outside the licensed indications for CARBOplatin in Ireland. Patients should be informed of the unlicensed nature of this indication and consented to treatment in line with the hospital's policy on the use of unlicensed medication and unlicensed or "off label" indications. Prescribers should be aware of their responsibility in communicating any relevant information to the patient and also in ensuring that the unlicensed or "off label" indication has been acknowledged by the hospital's Drugs and Therapeutics Committee, or equivalent, in line with hospital policy.

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